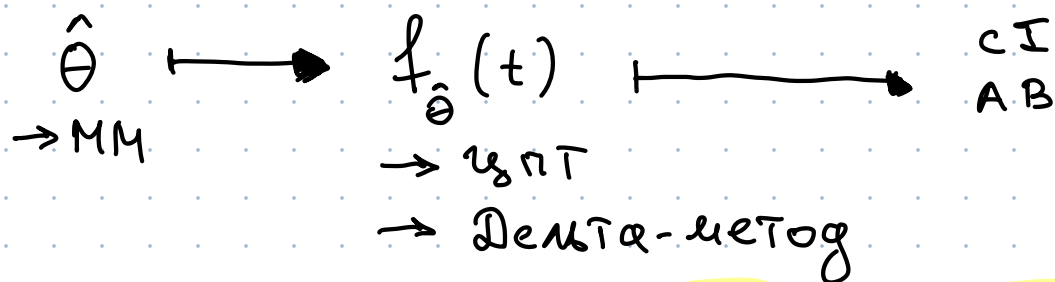


Больше асимптотики для Big Data

Что мы уже знаем:

Модель + Предположки: $X_1, \dots, X_n \sim \text{iid } \mathcal{F}_\theta$; $\sigma^2 < \infty$



$\bar{X}$	μ	P
$f(\bar{X})$	$\frac{1}{P}$	μ^2

σ^2	$\hat{\sigma}^2$	S^2
q_α	$\hat{q}_\alpha$	

$\text{cov}(X, Y)$
$\hat{\text{cov}}(X, Y)$

$$\bar{X}_n \overset{\text{asy}}{\sim} \mathcal{N}\left(\mu, \frac{\hat{\sigma}^2}{n}\right)$$

в следующем раз

через μ_4

$$S_n^2 \overset{\text{asy}}{\sim} \mathcal{N}\left(\sigma^2, \frac{\mu_4 - \sigma^4}{n}\right)$$

$$X_{(Ln+1)} \overset{\text{asy}}{\sim} \mathcal{N}\left(q_\alpha; \frac{\alpha \cdot (1-\alpha)}{n \cdot f(q_\alpha)^2}\right)$$

① Асимптотическое р-е. для $\hat{\sigma}^2$ и S^2

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \overline{x^2} - \bar{x}^2 - \text{смещенная оценка}$$

$$S_n^2 = \frac{n}{n-1} \cdot \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 - \text{несмещ. оценка}$$

$$f(\bar{x}^2, \bar{x}^2)$$

моменты

$$\mu_k = \mathbb{E}(X^k)$$

$$\mu_1 = \mu$$

$$\hat{\mu}_k = \overline{X^k} = \frac{1}{n} \sum_{i=1}^n X_i^k$$

$$\hat{\mu}_k \xrightarrow{P} \mu_k \quad \text{з.б.ч.}$$

сост. оценка

центральные моменты

$$\mu_k = \mathbb{E}([X - \mathbb{E}(X)]^k)$$

$$\mu_1 = 0$$

$$\mu_2 = \sigma^2 = \mu_2 - \mu_1^2$$

$$\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k$$

$$\tilde{\mu}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^k$$

$$\tilde{\mu}_k \xrightarrow{P} \mu_k \quad \text{з.б.ч.}$$

сост. оценка

$$\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k = \frac{1}{n} \sum_{i=1}^n ((X_i - \mu) - (\bar{X} - \mu))^k =$$

$$(a+b)^n = \sum_{k=0}^n C_n^k a^k b^{n-k}$$

не зав. от i

$$= \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^k C_k^j (X_i - \mu)^j (-1)^{k-j} (\bar{X} - \mu)^{k-j} =$$

$$= \sum_{j=0}^k C_k^j (-1)^{k-j} (\bar{X} - \mu)^{k-j} \cdot \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^j \right) \xrightarrow{P} C_k^j (-1)^{k-j} \cdot 1 \cdot \mu_j$$

$$\begin{aligned} & \downarrow \\ & 0^{k-j} = 0 \quad k > j \\ & 0^0 = 1 \quad k = j \end{aligned}$$

$$\tilde{\mu}_j \xrightarrow{P} \mu_j$$

$$\mu_k$$

$$\text{т.е. } \text{plim}_{n \rightarrow \infty} \hat{\mu}_k = \mu_k$$

и оценка сост.

ОпР.

T_n - оценка для θ

T_n - состоятельная, если

$$T_n \xrightarrow{P} \theta$$

T_n - асимпт. коррект. оценка, если

$$\sqrt{n} \cdot (T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \tau^2)$$

$$\sqrt{n} \cdot \frac{T_n - \theta}{\tau} \xrightarrow{d} \mathcal{N}(0, 1)$$

$U_n \xrightarrow{P} \tau$ - сост. оц. для τ

$$\sqrt{n} \frac{T_n - \theta}{U_n} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$S_n \xrightarrow{P} \sigma$$

$$\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{S_n} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$\sqrt{n}(S_n^2 - \sigma^2) \xrightarrow{d} ?$$

$$\sqrt{n}(\tilde{\mu}_2 - \mu_2) \xrightarrow{d} \mathcal{N}(0; \tau^2)$$

зпт

$$\tau^2 = ?$$

Haupteigenschaften τ^2 :

$$\begin{aligned}\tau^2 &= \text{Var}(\sqrt{n}(\tilde{\mu}_2 - \mu_2)) = n \cdot \text{Var}(\tilde{\mu}_2) = \\ &= n \cdot \frac{1}{n^2} \sum_{i=1}^n \text{Var}(x_i - \mu)^2 = \text{Var}(x_i - \mu)^2 = \\ &= \mathbb{E}(x_i - \mu)^4 - \mathbb{E}^2(x_i - \mu)^2 = \mu_4 - \mu_2^2 = \mu_4 - \sigma^4\end{aligned}$$

$$\tilde{\mu}_2 \stackrel{\text{asy}}{\sim} \mathcal{N}\left(\mu_2; \frac{\mu_4 - \sigma^4}{n}\right)$$

$$\begin{aligned}\hat{\sigma}_n^2 &= \frac{n-1}{n} \cdot S_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2 = \\ &\quad [(x_i - \mu) - (\bar{x}_n - \mu)] \\ &= \frac{1}{n} \sum_{i=1}^n \left[\underbrace{(x_i - \mu)^2}_{\tilde{\mu}_2} - 2(x_i - \mu)(\bar{x}_n - \mu) + \underbrace{(\bar{x}_n - \mu)^2} \right] = (*)\end{aligned}$$

$$\begin{aligned}-\frac{2}{n} \sum_{i=1}^n (x_i - \mu)(\bar{x}_n - \mu) &= -\frac{2(\bar{x}_n - \mu)}{n} \sum_{i=1}^n \underbrace{(x_i - \mu)}_{(\bar{x}_n - \mu)} = \\ &= -2(\bar{x}_n - \mu)^2\end{aligned}$$

$$(*) = \tilde{\mu}_2 - (\bar{x}_n - \mu)^2$$

$$\text{Korrig: } \sqrt{n}(S_n^2 - \sigma^2)$$

$$\frac{n-1}{n} S_n^2 - \frac{n-1}{n} \sigma^2 = \hat{\mu}_2 - (\bar{X}_n - \mu)^2 - \frac{n-1}{n} \sigma^2 \left(\times \sqrt{n} \cdot \frac{n}{n-1} \right)$$

$$\sqrt{n}(S_n^2 - \sigma^2) = \underbrace{\frac{n}{n-1}}_{\rightarrow 1} \sqrt{n} \hat{\mu}_2 - \sqrt{n} \mu_2 - \underbrace{\sqrt{n} \cdot \frac{n}{n-1} (\bar{X}_n - \mu)^2}_{\xrightarrow{d} 0}$$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

$$n(\bar{X}_n - \mu)^2 \xrightarrow{d} [\mathcal{N}(0, \sigma^2)]^2$$

$$\frac{\sqrt{n}}{n-1} \cdot W \xrightarrow{d} [\mathcal{N}(0, \sigma^2)]^2$$

$\rightarrow 0$

$$\sqrt{n}(S_n^2 - \sigma^2) \sim \sqrt{n}(\hat{\mu}_2 - \mu_2)$$

При $n \rightarrow \infty$ ведуть себя
одинаково

$$\mathcal{N}(0, \mu_4 - \sigma^4)$$

$$S_n^2 \overset{asy}{\sim} \mathcal{N}\left(\sigma^2; \frac{\mu_4 - \sigma^4}{n}\right)$$

$$S_n^2 \pm Z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{\mu}_4 - \hat{\sigma}^4}{n}}$$

② Стабилизация гусперсим

$$\frac{T_n - \theta}{u_n / \sqrt{n}} \stackrel{asy}{\sim} \mathcal{N}(0, 1)$$

$$\begin{array}{ccc} T_n & \xrightarrow{P} & \theta \\ u_n & \xrightarrow{P} & \sigma \end{array}$$

$$T_n \pm z_{1-\frac{\alpha}{2}} \cdot \frac{u_n}{\sqrt{n}}$$

$$\sqrt{n} \cdot (\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$$

$$\text{УПТ: } \hat{\theta}_n \stackrel{asy}{\sim} \mathcal{N}(\theta, \frac{\sigma^2(\theta)}{n})$$

$$\text{Дельта-метод: } g(\hat{\theta}_n) \stackrel{asy}{\sim} \mathcal{N}(g(\theta), \underbrace{\frac{\sigma^2(\theta)}{n} \cdot [g'(\theta)]^2}_{\text{хоруг} = 1/n})$$

хоруг = $1/n$

Опр. Стабилизация гусперсим

$$[g'(\theta)]^2 \sigma^2(\theta) = 1$$

$$g'(\theta) \cdot \sigma(\theta) = 1$$

$$g'(\theta) = \frac{1}{\sigma(\theta)} \quad g(\theta) = \int \frac{d\theta}{\sigma(\theta)}$$

$$g(\hat{\theta}) \pm z_{1-\frac{\alpha}{2}} \cdot \frac{1}{\sqrt{n}} \quad \text{Решаето отн. } \theta$$

Упражнения

Д. и. Вальга: $\hat{P}_n \pm Z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P}(1-\hat{P})}{n}}$

Д. и. Юнсока:

$$\frac{n}{n + Z_{1-\frac{\alpha}{2}}^2} \cdot \left(\left(\hat{P}_n + \frac{Z_{1-\frac{\alpha}{2}}^2}{2n} \right) \pm Z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{P}(1-\hat{P})}{n} + \frac{Z_{1-\frac{\alpha}{2}}^2}{4n}} \right)$$

Построим ещё один асимпт. г.ч. где гоеи:

$$\hat{P} \overset{asy}{\rightsquigarrow} \mathcal{K} \left(P; \frac{P(1-P)}{n} \right)$$

$$\sigma^2(P) = P(1-P)$$

$$g(\hat{P}) \overset{asy}{\rightsquigarrow} \mathcal{K} \left(g(P); \frac{1}{n} \right)$$

$$g(P) = \int \frac{dp}{\sqrt{P(1-P)}} = 2 \cdot \arcsin \sqrt{P}$$

$$2 \cdot \arcsin \sqrt{\hat{P}} \overset{asy}{\rightsquigarrow} \mathcal{K} \left(2 \arcsin \sqrt{P}, \frac{1}{n} \right)$$

$$2 \arcsin \sqrt{\hat{P}} - \frac{Z}{2\sqrt{n}} \leq 2 \arcsin \sqrt{P} \leq 2 \arcsin \sqrt{\hat{P}} + \frac{Z}{2\sqrt{n}}$$

$$\sin^2 \left(\arcsin \sqrt{\hat{P}} \pm \frac{Z_{1-\frac{\alpha}{2}}}{2\sqrt{n}} \right) - \text{гов. кт. г.ч. } P$$

Общий вид:
$$g^{-1}\left(g(\hat{\theta}_n) \pm \frac{Z_{1-\frac{\alpha}{2}}}{\sqrt{n}}\right)$$